



NYU SCHOOL OF PROFESSIONAL STUDIES

Bobcat

Agentic AI Recruiting Assistant

Building an AI-Powered System to Support Prospective Student Inquiries
and Reduce Faculty Administrative Burden

Faculty Time Is Being Wasted

A Fall 2025 needs assessment survey revealed a systemic challenge in prospective student inquiry management at NYU SPS.

The 4 Compounding Problems

- Faculty time diverted from teaching & research
- Inconsistent messaging confuses applicants
- No 24/7 availability across time zones
- No data on inquiry patterns for administrators

40%

of faculty contact prospective students multiple times / week

80%

report inconsistent information shared across faculty members

60%

of inquiries are routine & could be automated

Meet Bobcat

*An agentic AI recruiting assistant that handles the automatable 60% of inquiries —
freeing faculty to focus on what matters.*

 Program Matcher

 Academic Audit

 Deadline Tracker

 Admissions Guide

 Resume Analyzer

 Career Services

 Escalation Handler

24/7

Always Available

FERPA

Compliant by Design

Claude

Powered by Anthropic AI

How Bobcat Works

User Interface

React • NYU SPS branding • WCAG 2.1 AA • Embeddable widget

Agentic LLM Layer

Claude Sonnet 4 (Anthropic) • 7 Tool Agents • Tool-calling architecture

RAG Retrieval Engine

Pinecone Vector DB • OpenAI Embeddings (1536-dim) • Cosine similarity @ 0.75

Knowledge Base

54 faculty-approved Q&A pairs • 4 categories • FERPA-safe content

Tech Stack

- React.js (UI)
- FastAPI (Backend)
- Pinecone (VectorDB)
- Anthropic Claude API
- OpenAI Embeddings
- AWS / Heroku (Deploy)
- GitHub (DevOps)

Four Objectives. All Exceeded.

01

Knowledge Base

54 faculty-approved Q&A pairs (target: 50)
Sponsor sign-off Week 4

108%

of target

02

Prototype Built

React UI + FastAPI + Pinecone RAG
7 tool agents + WCAG 2.1 AA

88%

accuracy

03

Usability Study

7 faculty & admissions staff tested
Validation report sponsor-approved

4.1

/ 5.0 rating

04

Platform Handoff

Analytics dashboard, FERPA docs,
Tech architecture guide, Ops manual

298h

of 300 budget

The A/B Trial

4-week controlled experiment · April 6 – May 1, 2026

CONTROL

Manual Process

Faculty & staff respond
via email, phone, web forms

Response time: 24 hours
Consistency: Variable

VS

INTERVENTION

Bobcat AI

Agentic chatbot widget
embedded on SPS program page

Response time: < 10 seconds
Consistency: Guaranteed

PICO

Hypothesis

- P: Prospective grad students
- I: Bobcat chatbot
- C: Manual email/phone
- O: Speed · Accuracy · Satisfaction

Wk 1-2: Baseline (control only) → Wk 3-4: A/B Comparison → Final: Statistical Analysis

Every Target Exceeded.

Hypothesis: Fully Supported

88%

Response Accuracy
(Target: >85%)

<10s

First-Response Time
(Target: <1 min)

9%

Escalation Rate
(Target: <15%)

4.1/5

User Satisfaction
(Target: ≥3.5/5.0)

66%

Faculty Time Saved
(Target: >40%)

Bobcat in Action

 Bobcat — NYU SPS · 7 Agents Active

I have a finance background and know Python. Am I eligible for Fintech?

 Program Matcher ·  Academic Audit

Yes! Your finance background meets prerequisites. Python satisfies the programming requirement. Fintech is 36 credits, on-campus.

What's the Fall 2026 deadline?

 Deadline Tracker

Fall 2026 Regular Deadline: May 1, 2026. International students should apply by Priority 2 (Jan 15) for visa processing.

Smart Matching

Program recommendations based on background and career goals



Instant Response

Mean latency 4.3 seconds vs. 24-hour manual response time

Analytics Panel

Real-time dashboard: volume, accuracy, escalation, satisfaction

FERPA Safe

No SSN/NUID collected. Auto-escalation for sensitive queries

Grounded in Research

20 peer-reviewed papers · 15 primary references cited



30–50%

Workload Reduction

Huang & Zhang (2023) — AI chatbots reduce staff workload for routine inquiries while maintaining satisfaction. Our result: 66%.



82%

Resolution Rate (Benchmark)

Islam et al. (2024) — University chatbot at Semarang achieved 82% resolution. Bobcat achieved 88% on faculty-validated test set.



55 Studies

Systematic Review

Okonkwo & Ade-Ibijola (2021) — Administrative burden the #1 motivation for chatbot deployment in higher education globally.

Issues Encountered. All Resolved.

Knowledge Base Gaps

Resolved

Risk #2 (Prob: 3, Impact: 3)

Mid-project checkpoint Week 6. Added 5 Q&A pairs (49→54).
Flagging system implemented.

Faculty Strike / Sponsor Availability

Pending Sign-off

Risk #10 + Unplanned Event

All deliverables completed

Pinecone Retrieval Accuracy

Resolved

Risk #6 (Prob: 2, Impact: 3)

Raised similarity threshold 0.65→0.75. Reduced chunk size
512→256 tokens. Precision: 81%→94%.

WCAG 2.1 AA Violations

Resolved

Discovered in Week 9 Testing

3 violations fixed in 2 dev days. Contrast 3.8→5.2:1. aria-labels
added. Motion preferences respected.

The Path Forward

Phase 1 | ✓ COMPLETE

Capstone Pilot

- 54-question knowledge base
- 88% accuracy prototype
- 7-participant usability study
- Full documentation package

Phase 2 | RECOMMENDED

Controlled Student Pilot

- NYU IT security review + Anthropic DPA
- Bias testing (20 student profiles)
- Slate CRM integration
- 60-day live pilot on 1 program page

Phase 3 | LONG-TERM

Full Deployment

- Sitewide across 20+ SPS programs
- Replicate to NYU Steinhardt, Wagner, Tandon, Stern
- Multilingual support
- Proactive outreach & CRM analytics

WHAT'S NEXT

Demo